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Soil: Bottom soil may be loaded with freshwater. Microbes in water can convert the waste materials into usable forms like ammonia into nitrates which are used by the plants. Thus the soil fertility is maintained

Pesticides: In this system use of pesticides is avoided and hence it is eco-friendly.

Weeds: Since the plants are cultured in confined conditions, growth of weeds is completely absent. The utilization of nutrient by plants is high in this method

Artificial food for fishes: In this system plant waste and decays are utilized by fishes as food. So, the need for the use of supplementary feed can be minimized.

Fertilizer usage: Artificial or chemical fertilizers is not required for this system since the plants in the aquaponics utilize the nutrients from the fish wastes dissolved in water

Cultivable fishes like tilapia, trout, koi, gold fish, bass etc., are cultured in aquaponics. Common cultivable plants like tomato, pepper, lettuce, cucumber, and rose are co-cultivated in this method.

13.7 Aquaculture

Aquaculture has been practiced in varying forms for centuries dating to the time of the Phoenicians. India offers a huge potential for aquaculture development. Fish culture received notable attention in Tamil Nadu in 1911. Aquaculture is a branch of science that deals with the farming of aquatic organisms such as fish, molluscs, crustaceans and aquatic plants.

On the basis of source, aquaculture can be classified into three categories. They are (a) **Freshwater aquaculture** (b) **Brackish water aquaculture** (c) **Marine water aquaculture**. Culturing of fishes is called fish culture or pisciculture. Inland water bodies include freshwater bodies like rivers, canals, streams, lakes, flood plain wetlands, reservoirs, ponds, tanks and other derelict water bodies and ponds constructed for fresh water aquaculture. The pH of the freshwater should be around neutral and salinity below 5 ppt (parts per thousand).

Brackish water fishes spend most of its life in river mouths (estuaries) back waters, mangrove swamps and coastal lagoons. Estuarine fish are more common in Bengal and Kerala. Culturing of animals in the water having salinity range 0.5 2 30 ppt are called as brackish water culture. Fishes cultured in brackish water are Milk fish (*Chanos Chanos)*, Sea bass ('Koduva'), Grey mullet ('Madavai'), *Pearl spots ('Kari'meen)* etc,

Marine Fisheries deal with fishing operations along seacoasts. The Indian subcontinent approximately has a 5600 kms long coastline. About 80% of India's marine fish are supplied by the west coast and the remaining 20% by the east coast. The premier varieties are mackerels, sardines, sharks, and catfish. Marine edible fishes of Tamilnadu coast include both cartilaginous and bony fishes. Culturing of animals in the water salinity ranges from 30 - 35% is called Mariculture. Some fishes like *Chanos* sp, *Mugil cephalus* are cultured here. Culturing of animals in the salinity ranges from 36 - 40% is called Metahaline culture. Eg, Brine shrimp (*Artemia salina)*. Artemia is commonly known as the brine shrimp. It is a crustacean and